

Cultivating HeLa cells

Risk assessment

- Work with human-derived material (BSL-2)
- ▷ Wear gloves, safety glasses, lab coat
- Collect and dispose waste after inactivation as REGULATED MEDICAL WASTE



Reviewed: Mar 13, 2023

This is a bench card. Full protocol available online.

A > Splitting adherent HeLa cells

- R0081 0.06% Trypsin solution

- (1.) Bring the growth medium, PBS, and trypsin to room temperature.
- (2.) Aspirate off the growth medium and wash the cells once with PBS.
- (3.) Cover the plate with 0.05% trypsin/EDTA and incubate at 37 °C for 2–3 min.
- (4.) Add an equal volume of growth medium containing serum, or 1 × trypsin inhibitor solution if applicable. Pipette up and down. Squirt the surface of the dish to detach all adherent cells.

⌚ 2–3 min



Critical: Remove at least 90% of the adherent cells to avoid selection bias in the culture.



- (5.) *Optional:* Transfer the suspension to a conical tube. Centrifuge at $500 \times g$ for 5 min. Resuspend the pellet in fresh growth medium. Count cells and assess viability.
- (6.) Seed the cells at the desired density in a new culture dish. Spread by rocking the dish back and forth.

Critical: Do not swirl in a circular motion—this causes the cells to clump in the center of the dish.



F > Freezing HeLa cells for long-term storage

- Cryovial, polypropylene, non-pyrogenic, with seal rings
- Dimethyl sulfoxide

- (1.) Prepare cryo-preservation medium by supplementing DMEM with 20% FBS and 10% DMSO.
- (2.) Bring the cryo-preservation medium, PBS, and trypsin to room temperature.
- (3.) Detach cells from a confluent 10 cm dish using standard trypsinization.
- (4.) Collect the cells by centrifugation at $500 \times g$ for 3 min. Carefully aspirate the supernatant.
- (5.) Resuspend the pellet in 2 mL cryo-preservation medium to a final concentration of about $4.0 \times 10^6 \text{ mL}^{-1}$.
- (6.) Aliquot 1 mL per vial into two cryovials. Label with water-resistant marker or deep-freeze label.
- (7.) Slow-freeze the vials at -80°C overnight using a styrofoam container or freezing device. Transfer to liquid nitrogen storage the next day.



T > Thawing a cryo-preserved HeLa cell line

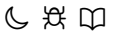
- (1.) Warm 20 mL of growth medium to 37 °C; aliquot 10 mL each into two conical tubes.
- (2.) **Critical:** Quickly thaw the cryogenic vial in a 37 °C water bath.

Critical: Ensure the lid is tightly sealed and keep the vial upright to avoid contamination. Wipe thoroughly with disinfectant before transferring to a sterile environment.



Cultivating HeLa cells

- (3.) Transfer the thawed cell suspension into one of the conical tubes with pre-warmed growth medium.
- (4.) Centrifuge at $500 \times g$ for 3 min. Carefully aspirate the supernatant to remove DMSO.
- (5.) Resuspend the cell pellet in the second tube containing 10 mL pre-warmed growth medium.
- (6.) Plate the resuspended cells onto a 10 cm culture dish.
- (7.) Incubate at 37°C under 5% CO_2 . Once attached and recovered, passage cells for the first time (P1).



S > **Bringing and maintaining HeLa S3 in suspension culture**

- | | |
|---|--|
| ☐ Magnetic spinner flask | ☐ Serum-free medium (EX-CELL® Medium) |
|---|--|

- (1.) Over the course of one to two weeks, gradually reduce the FBS content in the culture medium to 2%.
- (2.) Dissociate the cells using standard trypsinization.
- (3.) Plate the cells for 48 h in serum-free medium (without L-glutamine) such as EX-CELL® Medium.
- (4.) Dissociate the cells again. Count and assess viability.
- (5.) Seed $5.0 \times 10^5 \text{ mL}^{-1}$ viable cells into 3 magnetic spinner flasks. Stir gently at 40–70 rpm overnight.
- (6.) After 24 h, collect cells from the medium. Count and assess viability.



Critical: If viability is below 50%, use a Ficoll® gradient to remove dead cells by centrifugation.



- (7.) Pool suspension cells from two or three flasks into a new spinner flask with fresh medium. Inoculate at no more than $2.5 \times 10^5 \text{ mL}^{-1}$.
- (8.) Continue culture for three days with gentle stirring. Then harvest by centrifugation and check viability.



Critical: Do not disturb cells that have adhered to the spinner walls.



Recipe (available online) Troubleshooting (available online) Notes (available online)

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